

CRT Chapter 8

1. A has-a relationship means a class uses another class as a field (it “has” that object inside it), while an is-a relationship is shown when a class is a specialized version of another class through inheritance.
 2. For the second point, an object of the derived class can call both go() (inherited from the base class) and stop() (declared in the derived class).
 3. An abstract method in an abstract class is like a promise: every concrete subclass is required to provide its own implementation of that method. Overriding a method, on the other hand, is optional and is only done when a subclass wants to change or specialize the behavior of an existing concrete method. Abstract methods have no body in the abstract class, but an overriding method must include code.
 4. A class can extend only one abstract class, but it can implement many interfaces. Abstract classes are designed to be extended by related subclasses that share common state or behavior, whereas interfaces define capabilities that any unrelated class can choose to implement. Subclasses inherit from abstract classes, but they implement interfaces.
6. For the Wo, Bo, and Roo example:
- a) In Wo, doThat() is a public abstract method.
 - b) Wo is an interface that Roo agrees to implement.
 - c) Because Roo implements Wo, it is required to provide a concrete version of doThat().

- d) A Roo object has doThis() (from Roo), doNow() (from Bo), and doThat() (implementation of Wo), plus the usual methods from Object.
- e) Since Roo overrides doThis(), calling doThis() on a Roo object uses the Roo version (returning 10 instead of 2 from Bo).
- f) The call super(1) in Roo's constructor invokes Bo's constructor with the value 1, initializing the superclass field.
- g) The original doThis() in Bo can be reached from within Roo using super.doThis(), even though Roo overrides it.
- h) Yes, any method inside Roo can call Bo's version by writing super.doThis(), which will execute the implementation that returns 2.